

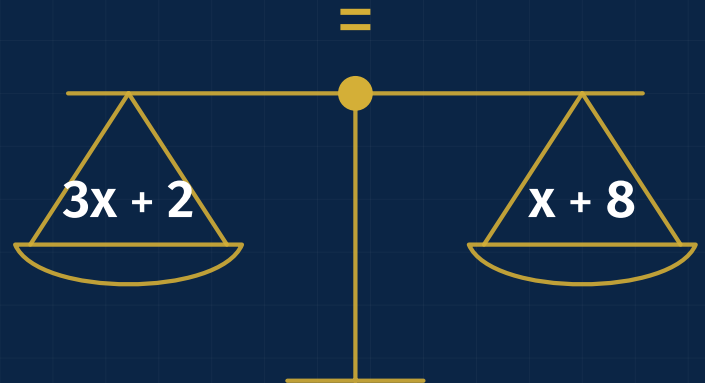


DR ESLAM AHMED
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CHAPTER 2 • ALGEBRA • TOPIC 5

Completing the Square

One structure. List four numbers, fill the blanks — and read the turning point straight off the answer.



6

VARIANTS

15

PAST-PAPER PARTS

40

PRACTICE QUESTIONS

YOU WILL MASTER

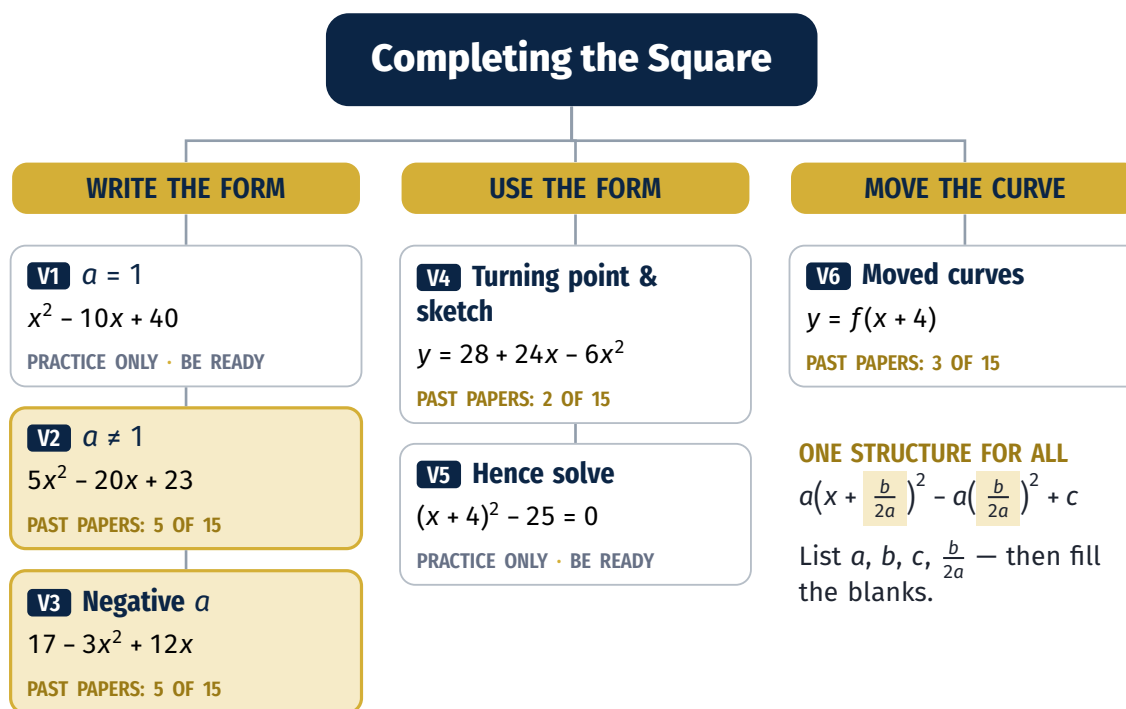
V1 $a = 1$ V2 $a \neq 1$ V3 Negative a

V4 Turning point & sketch V5 Hence solve V6 Moved curves

TOPIC MAP • EVERY VARIANT ON ONE PAGE

Completing the Square at a glance



Your notes


★ **How often does each variant come up?**

11 past-paper questions (Jan 2020 → Nov 2025) hold 15 parts; 40 practice parts — grouped by type.

	PAST PAPERS (OUT OF 15)	PRACTICE PARTS
V1 $a = 1$	0	17
V2 $a \neq 1$	5	6
V3 Negative a	5	5
V4 Turning point & sketch	2	9
V5 Hence solve	0	3
V6 Moved curves	3	0

WHAT THIS TELLS YOU The real papers never ask the easy $a = 1$ form — 10 of 15 parts have $a \neq 1$ or a negative a . That is why one structure for every a matters.

BEFORE YOU START

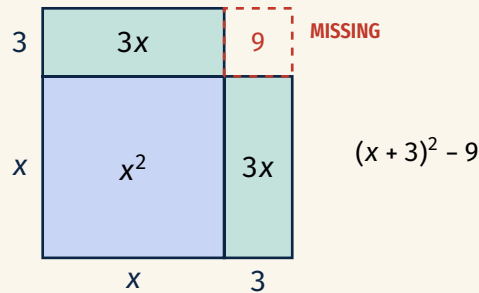
The big idea



Make a perfect square, then fix the extra bit.

$x^2 + 6x$ is **almost** the square $(x + 3)^2$:
put the $6x$ as two strips of $3x$ round the x^2 — one corner of $3 \times 3 = 9$ is missing.

$$\text{So } x^2 + 6x = (x + 3)^2 - 9.$$



Dr Eslam's structure — the same every time

STEP 1 • LIST YOUR ELEMENTS

$$a = ?$$

$$b = ?$$

$$c = ?$$

$$\frac{b}{2a} = ?$$

STEP 2 • USE THE STRUCTURE

$$a \left(x + \frac{b}{2a} \right)^2$$

keep it as it is

$$- a \left(\frac{b}{2a} \right)^2 + c$$

one number: use the calculator

? What the answer tells you

Turning point

$$a(x - p)^2 + q \rightarrow (p, q)$$

Flip the inside, keep the outside.

Min or max

$$a > 0 : \cup \quad a < 0 : \cap$$

Positive a : a minimum.
Negative a : a maximum.

Solve = 0

$$(x + 4)^2 = 25$$

Move $\rightarrow$ Root ($\pm$) $\rightarrow$ Solve.



CALCULATOR FIRST, THEN CHECK

Use the calculator for **the right part only**: $-a\left(\frac{b}{2a}\right)^2 + c$ is one number. Check your answer:

CALC the question and your form with $x = 2$ — same value?

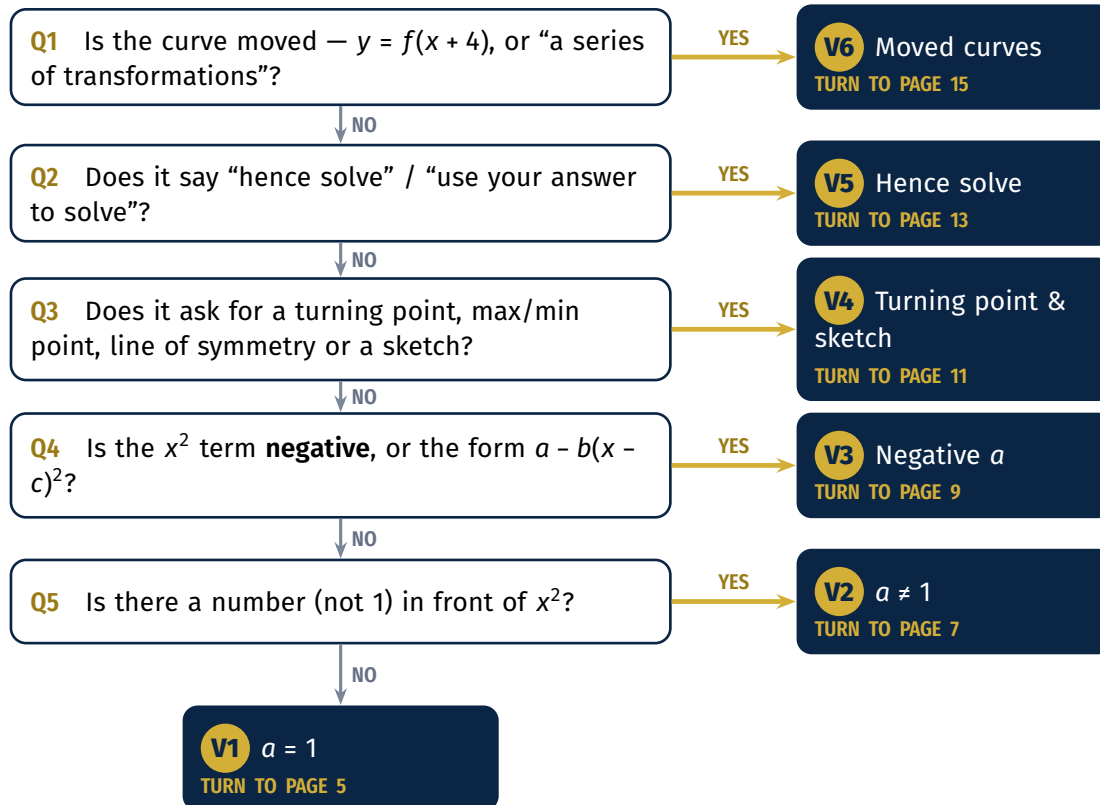
COUNT / RECOGNISE, THEN CHOOSE

Which variant am I?



Your notes

Read what the question wants, then walk down the ladder. The first **YES** tells you the variant — and the page to turn to.

**COLOUR CODE IN THE WORKING**

Blue = letter terms $4x$, $-3y$

Orange = plain numbers 17 , -13

STEP 1 **STEP 2** **STEP 3** **STEP 4** = which step of “How do I do it?” each line is.

ON EVERY VARIANT SPREAD

- | | | | |
|---|-----------------|---|-----------------------|
|  | What is it? |  | When you see → you do |
|  | How do I do it? |  | Watch out |
|  | Worked example |  | Exam tip |
|  | Check yourself |  | Calculator check |

V1

VARIANT 1

Completing the Square: $a = 1$ $x^2 + bx + c$ into $(x + p)^2 + q$ — the starting point for every variant.

SEEN IN PAST PAPERS

0 / 15

NOT ASKED YET • BE READY
PRACTICE QUESTIONS: 17

Your notes

? What is it?

IN PLAIN WORDS

Write $x^2 + bx + c$ as $(x + p)^2 + q$. The bracket number is **half of b** ($\frac{b}{2a}$ with $a = 1$). Then take its square away and add c : that is one number, q .

IT LOOKS LIKE

$x^2 + 6x - 7$

$x^2 - 10x + 40$

$x^2 + 4x - 9$

$$x^2 - 10x + 40 \longrightarrow (x - 5)^2 - (-5)^2 + 40 \longrightarrow (x - 5)^2 + 15$$

HALF OF -10

-25 + 40

Half of b goes in the bracket; its square comes back out.
 How do I do it?

1 List

$a = 1, b, c, \frac{b}{2a} = \frac{b}{2}.$

2 Structure

$(x + \frac{b}{2})^2 - (\frac{b}{2})^2 + c$ — the same number in both brackets.

3 Right part

$-(\frac{b}{2})^2 + c$ is one number: calculator.

4 Match

Read the form: $(x + a)^2 + b \rightarrow$ say what a and b are.

LIST, THEN STRUCTURE Half of b in the bracket, minus its square.

V1 Completing the Square: $a = 1$

VARIANT 1 · SEE IT DONE, THEN TRY

 Your notes
 WHEN YOU SEE → YOU DO

"Write ... in the form $(x + a)^2 + b$ " → Complete the square: half of b

"Find the value of a and b " → Your answer's numbers — match them to the form

" $x^2 - 12x + a = (x + b)^2$ " → No q left: $a = \left(\frac{-12}{2}\right)^2 = 36$

 WORKED EXAMPLE **PRACTICE**

Completing the Square · Medium Q3 · 2 marks

Express $x^2 - 10x + 40$ in the form $(x + a)^2 + b$, where a and b are integers.

STEP 1 $a = 1, b = -10, c = 40, \frac{b}{2a} = -5$ ◀ list your elements

STEP 2 $(x - 5)^2 - (-5)^2 + 40$ ◀ structure

STEP 3 $(x - 5)^2 - 25 + 40 = (x - 5)^2 + 15$ ◀ right part: one number

STEP 4 the form's $a = -5$, the form's $b = 15$ ◀ match the letters

ANSWER $(x - 5)^2 + 15$ ✓ CALC $x = 2$: both sides 24 WATCH OUT✗ $(x - 5)^2 + 40$ ✓ $(x - 5)^2 - 25 + 40$

$(x - 5)^2$ gives an extra +25 — you must take it away.

 EXAM TIP

The letters in the form are not your a, b, c . In $(x + a)^2 + b$ here $a = -5$ and $b = 15$.

 Check yourself

calculator first · answers on page 17

1.1 Practice · MediumWrite $x^2 + 6x - 7$ in the form $(x + a)^2 + b$

Answer:

1.2 Practice · MediumWrite $x^2 - 10x + 22$ in the form $(x - a)^2 - b$

Answer:

1.3 Practice · Hard
 $x^2 - 12x + a = (x + b)^2$.
Find a and b .

Answer:

V2**VARIANT 2** MOST COMMON IN THE EXAM

Completing the Square: $a \neq 1$

A number in front of x^2 — the same structure, with a in both places.

SEEN IN PAST PAPERS

5 / 15

PRACTICE QUESTIONS: 6



Your notes

Came up in: Jan 2022 P1HR Q19(b) • Jun 2022 P1HR Q23(a) • Jun 2023 P2H Q19 • May 2024 P1HR Q20(a) • Nov 2025 P1H Q21

? What is it?

In plain words

Now $ax^2 + bx + c$ goes into $a(x + p)^2 + q$.

Use **the same structure**: the bracket number is $\frac{b}{2a}$, and the a appears **twice** — in front of both brackets.

$\frac{b}{2a}$ can be a fraction ($-\frac{11}{4}$) — keep it exact.

$$5x^2 - 20x + 23$$

$$\frac{b}{2a} = \frac{-20}{10} = -2$$

$$5(x-2)^2 - 5(-2)^2 + 23$$

CALCULATOR: $-20 + 23 = 3$

IT LOOKS LIKE

$$5x^2 - 20x + 23$$

$$3x^2 + 12x + 19$$

$$2x^2 - 11x + 9$$

$$5(x - 2)^2 + 3$$

How do I do it?

1 List

a , b , c and $\frac{b}{2a}$ (divide by $2a$, not by 2).

2 Structure

$$a\left(x + \frac{b}{2a}\right)^2 - a\left(\frac{b}{2a}\right)^2 + c.$$

3 Right part

$-a\left(\frac{b}{2a}\right)^2 + c$ on the calculator — one number.

4 Match

Read off the letters the form asks for.

THE a GOES IN BOTH PLACES $-a\left(\frac{b}{2a}\right)^2$, not $-\left(\frac{b}{2a}\right)^2$.

V2 Completing the Square: $a \neq 1$

VARIANT 2 · SEE IT DONE, THEN TRY

 Your notes
 WHEN YOU SEE → YOU DO

“Express $5x^2 - 20x + 23$ in the form $a(x - b)^2 + c$ ”

→ List → structure with a twice

“where a , b and c are integers”

→ Match the form's letters to your numbers

“numbers to be found” (not integers)

→ $\frac{b}{2a}$ may be a fraction — keep it WORKED EXAMPLE **PAST PAPER**

November 2025 · Paper 1H · Q21 · 3 marks

Express $5x^2 - 20x + 23$ in the form $a(x - b)^2 + c$ where a , b and c are integers.

STEP 1 $a = 5$, $b = -20$, $c = 23$, $\frac{b}{2a} = \frac{-20}{10} = -2$ ◀ list your elements

STEP 2 $5(x - 2)^2 - 5(-2)^2 + 23$ ◀ structure: a in both places

STEP 3 $5(x - 2)^2 - 20 + 23$ ◀ right part on the calculator

STEP 4 $5(x - 2)^2 + 3$ ◀ the form: $a = 5$, $b = 2$, $c = 3$

ANSWER $5(x - 2)^2 + 3$ ✓ CALC $x = 2$: both sides 3 WATCH OUT

$$\times 5(x - 2)^2 - (-2)^2 + 23 = 5(x - 2)^2 + 19$$

$$\checkmark 5(x - 2)^2 - 5(-2)^2 + 23 = 5(x - 2)^2 + 3$$

The a multiplies the second bracket too — Dr Eslam's most-seen slip.

 EXAM TIP

5 of the 15 past-paper parts are this variant, 3 marks each. Writing the list line earns you the first mark.

 Check yourself

calculator first · answers on page 17

2.1 Jan 2022 P1HR

Express $3x^2 + 12x + 19$ in the form $a(x + b)^2 + c$

Answer:

2.2 Jun 2023 P2H

Express $3x^2 - 6x + 5$ in the form $a(x - b)^2 + c$

Answer:

2.3 Practice · Hard

Express $4x^2 - 8x + 7$ in the form $a(x + b)^2 + c$

Answer:

V3**VARIANT 3** MOST COMMON IN THE EXAM**Negative a**

The x^2 is negative and written last.
Reorder, same structure, number first.

SEEN IN PAST PAPERS

5 / 15

PRACTICE QUESTIONS: 5


Your notes

Came up in: Jan 2020 P2HR Q23 • Jan 2022 P1H Q20(a) • Jun 2022 P1H Q24 • May 2024 P1H Q25 • Jun 2025 P2H Q25(a)

? What is it?**IN PLAIN WORDS**

The quadratic is written backwards with a **negative x^2** : $17 - 3x^2 + 12x$. **Reorder** it to $-3x^2 + 12x + 17$ (so $a = -3$), use the **same structure** — the signs look after themselves — then write the answer **number first**: $29 - 3(x - 2)^2$.

IT LOOKS LIKE

$17 - 3x^2 + 12x$

$7 - 12x - 2x^2$

$28 + 24x - 6x^2$

1 REORDER

$-3x^2 + 12x + 17$

2 STRUCTURE

$-3(x - 2)^2 - (-3)(-2)^2 + 17$

$-(-3)(4) = +12, 12 + 17 = 29$

3 NUMBER FIRST

$29 - 3(x - 2)^2$

Reorder → structure → number first. Watch $-a$: with $a = -3$ it becomes $+3$.

**How do I do it?****1 Reorder**

Write it as $ax^2 + bx + c$: here a is negative.

2 List

$a, b, c, \frac{b}{2a}$ — the sign of a included.

3 Structure

$a(x + \frac{b}{2a})^2 - a(\frac{b}{2a})^2 + c$; the calculator does the right part.

4 Number first

Rewrite in the form asked: $p - q(x - r)^2$.

REORDER FIRST The structure never changes — only the signs inside it.

V3 Negative a

VARIANT 3 • SEE IT DONE, THEN TRY


 Your notes
WHEN YOU SEE → YOU DO

"in the form $a - b(x - c)^2$ " or
 " $a + b(x + c)^2$ "

→ **Negative a: reorder, structure, number first**

$28 + 24x - 6x^2$ (written backwards)

→ **Reorder:** $-6x^2 + 24x + 28$

"in terms of q": $q + 12x - qx^2$

→ **Same structure with letters:** $a = -q$

WORKED EXAMPLE **PAST PAPER**

May 2024 • Paper 1H • Q25 • 4 marks

$f(x) = 17 - 3x^2 + 12x$. Write $f(x)$ in the form $a - b(x - c)^2$ where a , b and c are constants.

STEP 1 $-3x^2 + 12x + 17$

◀ reorder

STEP 2 $a = -3$, $b = 12$, $c = 17$, $\frac{b}{2a} = \frac{12}{-6} = -2$

◀ list (sign of a in)

STEP 3 $-3(x - 2)^2 - (-3)(-2)^2 + 17 = -3(x - 2)^2 + 29$

◀ structure; right part = $12 + 17$

STEP 4 $29 - 3(x - 2)^2$

◀ number first: $a = 29$, $b = 3$,
 $c = 2$

ANSWER $29 - 3(x - 2)^2$

✓ CALC $x = 0$: both sides 17

WATCH OUT

✗ $-3(x - 2)^2 - 12 + 17$

✓ $-3(x - 2)^2 + 12 + 17$

$-a(\frac{b}{2a})^2$ with $a = -3$: $-(-3)(4) = +12$.
 Minus times minus.

EXAM TIP

Write the final line in the exact order of the form: $29 - 3(x - 2)^2$, not $-3(x - 2)^2 + 29$.

Check yourself

calculator first • answers on page 17

3.1 Jan 2020 P2HR

Express $7 - 12x - 2x^2$
 in the form $a + b(x + c)^2$

Answer:

3.2 Practice • Very hard

Express $7 - 4x - x^2$ in
 the form $p - (x + q)^2$

Answer:

3.3 Practice • Very hard

Express $5 + 6x - x^2$ in
 the form $p - (x - q)^2$

Answer:

V4

VARIANT 4

Turning Point & Sketch

The completed square shows the lowest (or highest) point of the curve.

SEEN IN PAST PAPERS

2 / 15

PRACTICE QUESTIONS: 9



Your notes

Came up in: Jan 2022 P1H Q20(b) · Jun 2025 P2H Q25(b)

? What is it?**In plain words**

In $y = a(x - p)^2 + q$ the square is never negative, so the curve turns at $x = p$, height q : the **turning point is (p, q)** .

Flip the inside, keep the outside:

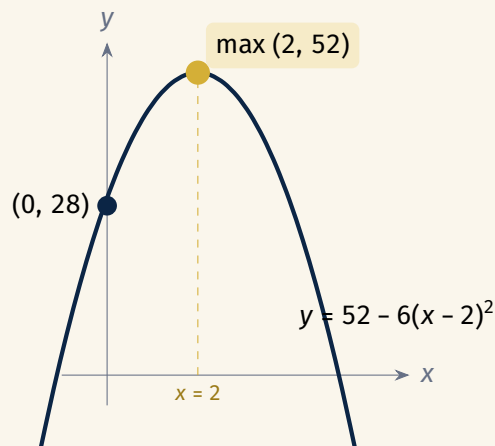
$(x + 7)^2 + 3 \rightarrow (-7, 3)$.

$a > 0$: a $\cup$ with a minimum. $a < 0$: a $\cap$ with a maximum.

IT LOOKS LIKE

$$y = (x + 3)^2 + 5$$

$$y = 52 - 6(x - 2)^2$$

**How do I do it?****1 Complete the square**

Use V1–V3 to get $a(x - p)^2 + q$.

2 Turning point

Flip the inside, keep the outside: (p, q) .

3 Min or max

$a > 0 \rightarrow \cup$, minimum. $a < 0 \rightarrow \cap$, maximum.

4 Sketch

Three things only: the shape, the turning point, the y-cut $(0, c)$.

FLIP THE INSIDE, KEEP THE OUTSIDE

Line of symmetry: $x =$ the flipped inside.

V4 Turning Point & Sketch

VARIANT 4 • SEE IT DONE, THEN TRY


 Your notes
WHEN YOU SEE → YOU DO

“turning point”, “minimum point”,
“maximum point”

→ **Flip the inside, keep the outside**

“line of symmetry”

→ **$x =$ the flipped inside number**

“Sketch ... show the turning point and
the y -intercept”

→ **Shape + turning point + $(0, c)$** **WORKED EXAMPLE** **PAST PAPER**

June 2025 • Paper 2H • Q25 • 6 marks

(a) Write $28 + 24x - 6x^2$ as $a - b(x - c)^2$. (b) Sketch $y = 28 + 24x - 6x^2$:
turning point and y -intercept.

STEP 1 $-6x^2 + 24x + 28 : \frac{b}{2a} = \frac{24}{-12} = -2$ ◀ (a) reorder and list

STEP 1 $-6(x - 2)^2 + 24 + 28 = 52 - 6(x - 2)^2$ ◀ (a) structure, number first

STEP 2+3 $\max(2, 52), a < 0 : \cap$ ◀ (b) flip inside, keep outside;
the shape

STEP 4 $x = 0 : y = 28 \rightarrow (0, 28)$ ◀ (b) the y -cut

ANSWER $52 - 6(x - 2)^2; \max(2, 52), (0, 28)$

✓ sketch: page A

WATCH OUT

✗ $(x + 7)^2 + 3 \rightarrow$ turning point $(7, 3)$

✓ $(x + 7)^2 + 3 \rightarrow (-7, 3)$

The inside number flips sign; the outside number stays.

EXAM TIP

A sketch earns marks for exactly three things — shape, turning point, y -cut. Label both points with coordinates.

Check yourself

calculator first • answers on page 17

4.1 Practice • Medium

Turning point of
 $y = (x + 3)^2 + 5$

Answer:

4.2 Practice • Hard

Turning point of
 $y = x^2 + 14x + 52$

Answer:

4.3 Practice • Hard

Line of symmetry of
 $y = 3(x - 2)^2 - 5$

Answer:

V5

VARIANT 5

Hence Solve

Use the completed square to solve = 0:
Move → Root → Solve.

SEEN IN PAST PAPERS

0 / 15

NOT ASKED YET • BE READY
PRACTICE QUESTIONS: 3


Your notes
? What is it?**In plain words**

Once you have $(x + 4)^2 - 25 = 0$, the letter appears **only once**. Undo it: **move** the number, take the **root** (two answers, $\pm$), **solve**.

If the root is not exact, leave a **surd**:
 $x = -2 \pm 2\sqrt{5}$.

IT LOOKS LIKE

$x^2 + 8x - 9 = 0$

$x^2 + 4x - 16 = 0$

MOVE

$(x + 4)^2 = 25$

ROOT $\pm$

$x + 4 = \pm 5$

SOLVE

$x = 1, -9$

How do I do it?**1 Complete the square**

$x^2 + 8x - 9 = (x + 4)^2 - 25 \text{ (V1)}.$

2 Move

Move the number: $(x + 4)^2 = 25$.

3 Root

Square root **both** ways: $x + 4 = \pm 5$ (or $\pm\sqrt{20}$).

4 Solve

$x = -4 + 5 = 1$ or $x = -4 - 5 = -9$. Simplify surds.

MOVE → ROOT → SOLVE Dr Eslam's three moves.

V5 Hence Solve

VARIANT 5 · SEE IT DONE, THEN TRY



Your notes
 WHEN YOU SEE → YOU DO

"Use your answer to part (i) to solve" → **Move → Root → Solve**

"Give your answers in surd form" → **Keep $\sqrt{}$, simplify: $\sqrt{20} = 2\sqrt{5}$**

" $x = \dots$ or $x = \dots$ " → **Two answers — the $\pm$**

 WORKED EXAMPLE PRACTICE

Completing the Square · Hard Q10 · 4 marks

(i) Write $x^2 + 8x - 9$ in the form $(x + k)^2 + h$. (ii) Use part (i) to solve $x^2 + 8x - 9 = 0$.

STEP 1 $(x + 4)^2 - 16 - 9 = (x + 4)^2 - 25$ ◀ (i) half of 8 is 4

STEP 2 $(x + 4)^2 = 25$ ◀ (ii) move

STEP 3 $x + 4 = \pm 5$ ◀ root, both signs

STEP 4 $x = 1$ or $x = -9$ ◀ solve

ANSWER (i) $(x + 4)^2 - 25$ (ii) $x = 1$ or $x = -9$ ✓ calculator roots: 1, -9

 WATCH OUT

✗ $(x + 4)^2 = 25 \Rightarrow x = 1$ only

✓ $x + 4 = \pm 5 \Rightarrow x = 1$ or $x = -9$

A square root has two answers.
Forgetting -5 loses the last mark.

 EXAM TIP

The calculator's roots must match your answers — and in surd form, $-2 \pm 2\sqrt{5}$ gives 2.47 and -6.47.

 Check yourself

calculator first · answers on page 17

5.1 Practice · Hard

$x^2 + 4x - 16 = (x + 2)^2 - 20$.
Solve $x^2 + 4x - 16 = 0$ in surd form.

Answer:

5.2 Practice · Hard

$x^2 - 18x - 27 =$
 $(x - 9)^2 - 108$. Solve
 $x^2 - 18x - 27 = 0$.

Answer:

5.3 Dr Eslam

Write $x^2 - 4x - 5$ as
 $(x - p)^2 - q$, hence solve
 $x^2 - 4x - 5 = 0$.

Answer:

V6

VARIANT 6

Moved Curves

$y = f(x + 4)$ or a list of transformations:
move the turning point.

SEEN IN PAST PAPERS

3 / 15

PRACTICE QUESTIONS: 0


Your notes

Came up in: May 2021 P2H Q22 · Jun 2022 P1HR Q23(b) · May 2024 P1HR Q20(b)

? What is it?**In plain words**

Find the turning point of $y = f(x)$ by completing the square. Then:

■ $y = f(x + 4)$ moves the curve **4 left** (inside: opposite way).

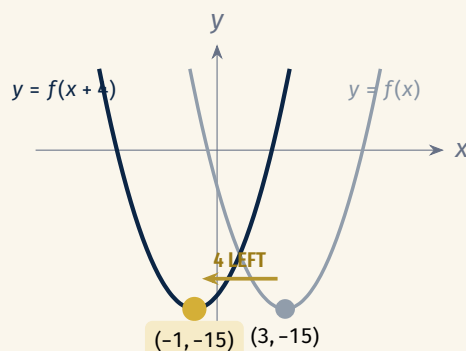
■ $y = f(x) + 3$ moves it **3 up** (outside: same way).

Spot $f(x + 4)$ hiding in

$2(x + 4)^2 - 12(x + 4) + 3$: every x became $(x + 4)$.

IT LOOKS LIKE

$$y = 2(x + 4)^2 - 12(x + 4) + 3$$

**How do I do it?****1 Turning point of f**

Complete the square: $f(x) = 2(x - 3)^2 - 15 \rightarrow (3, -15)$.

2 Inside change

$f(x + k)$: move k **left** — subtract k from the x -coordinate.

3 Outside change

$f(x) + k$: move k **up** — add k to the y -coordinate.

4 Describe

From $y = x^2$: translate $\begin{pmatrix} p \\ 0 \end{pmatrix}$, stretch $\parallel$ y -axis s.f. a , translate $\begin{pmatrix} 0 \\ q \end{pmatrix}$.

INSIDE: OPPOSITE · OUTSIDE: SAME

Links to Topic: Transformations of Graphs.

V6 Moved Curves

VARIANT 6 • SEE IT DONE, THEN TRY



Your notes
WHEN YOU SEE → YOU DO

$$y = 2(x + 4)^2 - 12(x + 4) + 3$$

→ It is $f(x + 4)$: the turning point moves 4 left“ $y = f(x - 3)$, find the minimum point”→ Turning point of f , then 3 right

“describe fully a series of transformations”

→ Translate • stretch • translate

WORKED EXAMPLE **PAST PAPER**

June 2022 • Paper 1HR • Q23 • 5 marks

(a) Express $2x^2 - 12x + 3$ in the form $a(x + b)^2 + c$. (b) **C**: $y = 2(x + 4)^2 - 12(x + 4) + 3$. Find the minimum point **M** of **C**.

STEP 1 $a = 2, b = -12, c = 3, \frac{b}{2a} = -3$ ◀ (a) list

STEP 1 $2(x - 3)^2 - 2(-3)^2 + 3 = 2(x - 3)^2 - 15$ ◀ (a) structure

STEP 2 $f(x) = 2x^2 - 12x + 3$: minimum (3, -15) ◀ (b) turning point of f

STEP 3 **C** is $y = f(x + 4)$: $3 - 4 = -1$ ◀ (b) inside +4: 4 left

ANSWER (a) $2(x - 3)^2 - 15$ (b) **M**(-1, -15) ✓ $x = -1$: $2(9) - 12(3) + 3 = -15$

WATCH OUT× $f(x + 4) \rightarrow 4$ right: (7, -15)✓ $f(x + 4) \rightarrow 4$ left: (-1, -15)

A change inside the bracket moves the curve the opposite way.

EXAM TIP

Check by substituting your x into the curve: $x = -1$ gives $y = -15$, the lowest value.

Check yourself

calculator first • answers on page 17

6.1 May 2024 P1HR

$2x^2 - 11x + 9 = 2\left(x - \frac{11}{4}\right)^2 - \frac{49}{8}$. Find the minimum point of $y = 2(x - 3)^2 - 11(x - 3) + 9$.

Answer:

6.2 Dr Eslam

$f(x) = (x - 2)^2 - 9$. Write down the turning point of $y = f(x + 3)$.

Answer:

6.3 May 2021 P2H

$g(x) = 2x^2 - 12x + 13 = 2(x - 3)^2 - 5$. Describe how $y = x^2$ maps onto $y = g(x)$.

Answer:

CHECK YOURSELF

Answers

Every answer below was checked with sympy (check_answers.py). Got one wrong? Go back to that variant's Watch-out box first.

1.1 $(x + 3)^2 - 16$

1.2 $(x - 5)^2 - 3$

1.3 $a = 36, b = -6$

2.1 $3(x + 2)^2 + 7$

2.2 $3(x - 1)^2 + 2$

2.3 $4(x - 1)^2 + 3$

3.1 $25 - 2(x + 3)^2$

3.2 $11 - (x + 2)^2$

3.3 $14 - (x - 3)^2$

4.1 $(-3, 5)$

4.2 $(-7, 3)$

4.3 $x = 2$

5.1 $x = -2 \pm 2\sqrt{5}$

5.2 $x = 9 \pm 6\sqrt{3}$

5.3 $x = 5$ or $x = -1$

6.1 $\left(\frac{23}{4}, -\frac{49}{8}\right)$

6.2 $(-1, -9)$

6.3 translate $\begin{pmatrix} 3 \\ 0 \end{pmatrix}$,
stretch $\parallel$ y-axis s.f. 2,
translate $\begin{pmatrix} 0 \\ -5 \end{pmatrix}$

THE WHOLE TOPIC ON ONE PAGE

Completing the Square cheat sheet



Your notes
LIST $a, b, c, \frac{b}{2a} \rightarrow$ STRUCTURE

$$a\left(x + \frac{b}{2a}\right)^2 - a\left(\frac{b}{2a}\right)^2 + c$$

V1 $a = 1$

$$x^2 - 10x + 40 = (x - 5)^2 + 15$$

Half of b in the bracket, minus its square.**V2** $a \neq 1$ **5/15**

$$5x^2 - 20x + 23 = 5(x - 2)^2 + 3$$

The a goes in both places.**V3** Negative a **5/15**

$$17 - 3x^2 + 12x = 29 - 3(x - 2)^2$$

Reorder, structure, number first.

V4 Turning point **2/15**

$$52 - 6(x - 2)^2 \rightarrow \max(2, 52)$$

Flip the inside, keep the outside. Sketch: shape, TP, $(0, c)$.**V5** Hence solve

$$(x + 4)^2 = 25 \Rightarrow x = 1, -9$$

Move $\rightarrow$ Root $(\pm) \rightarrow$ Solve.**V6** Moved curves **3/15**

$$f(x + 4) : 4 \text{ left}$$

Inside: opposite way. Outside: same way.

**TOP 3 LOST MARKS**

1. $-(\frac{b}{2a})^2$ without the $a \times$ 2. $(x + 7)^2 + 3 \rightarrow (7, 3) \times$ 3. $\sqrt{25} = 5$ only, no $\pm \times$

BOTTOM LINE List $\rightarrow$ structure $\rightarrow$ calculator for the right part · flip the inside, keep the outside.